Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The foot of the perpendicular from $(0, 2, 3)$ to the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$ is

(A) $(-2, 3, 4)$

(B) $(2, -1, 3)$

(C) $(2, 3, -1)$

(D) $(3, 2, -1)$

Q.2

Equation of plane which contains the line $\frac{x-1}{1} = \frac{y-2}{3} = \frac{z-3}{2}$ and which is perpendicular to the plane $2x + 7y + 5z = 2$ is

(A) $x + y + z = 6$

(B) $-x + y + z = 2$

(C) $2x - y + z = 3$

(D) $x - y + z = 2$

Q.3

The equation of the plane containing the line $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ is $A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$, where

(A) $Ax_1 + By_1 + Cz_1 = 0$

(B) $Al + Bm + Cn = 0$

(C) $A/l = B/m = C/n$

(D) $lx_1 + my_1 + nz_1 = 0$

Q.4

The lines and $\vec{r} = \vec{a} + \lambda(\vec{b} \times \vec{c})$ and $\vec{r} = \vec{b} + \vec{\mu}(\vec{c} \times \vec{a})$ will intersect if

(A) $\vec{a} \times \vec{c} = \vec{b} \times \vec{c}$

(B) $\vec{a}.\vec{c} = \vec{b} \cdot \vec{c}$

(C) $\vec{b} \times \vec{a} = \vec{c} \times \vec{a}$

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(D) none of these

Q.5

If l_1, m_1, n_1 and l_2, m_2, n_2 are DCs of the two lines inclined to each other at an angle θ, then the DCs of the bisector of the angle between these lines are

(A) $\frac{l_1+l_2}{2\sin\theta/2}, \frac{m_1+m_2}{2\sin\theta/2}, \frac{n_1+n_2}{2\sin\theta/2}$

(B) $\frac{l_1+l_2}{2\cos\theta/2}, \frac{m_1+m_2}{2\cos\theta/2}, \frac{n_1+n_2}{2\cos\theta/2}$

(C) $\frac{l_1-l_2}{2\sin\theta/2}, \frac{m_1-m_2}{2\sin\theta/2}, \frac{n_1-n_2}{2\sin\theta/2}$

(D) $\frac{l_1-l_2}{2\cot\theta/2}, \frac{m_1-m_2}{2\cot\theta/2}, \frac{n_1-n_2}{2\cot\theta/2}$

Q.6

The projection of the vector $\hat{i}+\hat{j}+\hat{k}$ on the line whose vector equation is $\vec{r}=(3+t)\hat{i}+(2t-1)\hat{j}+3t\hat{k}, t$ being the scalar parameter, is

(A) $\frac{1}{\sqrt{14}}$

(B) 0

(C) $\frac{6}{\sqrt{14}}$

(D) none of these

Q.7

Volume of tetrahedron formed by the planes $x+y=0, y+z=0, z+x=0, x+y+z-1=0$ is

(A) $\frac{1}{6}$

(B) $\frac{1}{3}$

(C) $\frac{2}{3}$

(D) none of these

Q.8

Let L be the line of intersection of the planes $2x+3y+z=1$ and $x+3y+2z=2$. If L makes an angle α with the positive x-axis, then $\cos\alpha$ equals

(A) $\frac{1}{\sqrt{3}}$

(B) $\frac{1}{2}$

(C) 1

(D) $\frac{1}{\sqrt{2}}$

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Q.9

If O is the origin and $OP = 3$ with direction ratios $-1, 2, -2$, then coordinate of P is

(A) $(1, 2, 2)$

(B) $(-1, 2, -2)$

(C) $(-3, 6, -9)$

(D) $\left(-\frac{1}{3}, \frac{2}{3}, -\frac{2}{3}\right)$

Q.10

The plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 3$ meets the coordinate axes in A, B, C. The centroid of the triangle ABC is

(A) $\left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$

(B) $\left(\frac{3}{a}, \frac{3}{b}, \frac{3}{c}\right)$

(C) $\left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right)$

(D) (a, b, c)

Q.11

The three planes $x + ay + (b + c)z + d = 0, x + by + (c + a)z + d = 0$ and $x + cy + (a + b)z + d = 0$

(A) intersect in a point

(B) intersect in a line

(C) intersect along different lines

(D) are mutually perpendicular

Q.12

The distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$, is

(A) 1

(B) $\frac{6}{7}$

(C) $\frac{7}{6}$

(D) none of these

Q.13

The equation of the plane through the points $(2, 2, 1)$ and $(9, 3, 6)$ and $\perp$ to the plane $2x + 6y + 6z - 1 = 0$ is

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(A) $3x + 4y + 5z = 9$

(B) $3x + 4y - 5z = 9$

(C) $3x + 4y - 5z = -9$

(D) none of these

Q.14

Let $A(1, -1, 1)$ and $B(-1, 1, -1)$ be the vertices of ΔABC, such that $\angle A = \angle B$. The locus of the vertex C is

(A) $x + 2y + z = 0$

(B) $2x + y - z = 0$

(C) $x - y + z = 0$

(D) $x - 2y + z = 0$

Q.15

The angle between two diagonals of a cube will be

(A) $\sin^{-1} \frac{1}{3}$

(B) $\cos^{-1} \frac{1}{3}$

(C) variable

(D) none of these

Q.16

The vector equation of the plane through the point $2\hat{i} - \hat{j} - 4\hat{k}$ and parallel to the plane $\vec{r} \cdot (4\hat{i} - 12\hat{j} - 3\hat{k}) - 7 = 0$ is

(A) $\vec{r} \cdot (4\hat{i} - 12\hat{j} - 3\hat{k}) = 0$

(B) $\vec{r} \cdot (4\hat{i} - 12\hat{j} - 3\hat{k}) = 32$

(C) $\vec{r} \cdot (4\hat{i} - 12\hat{j} - 3\hat{k}) = 12$

(D) none of these

Q.17

If the angle θ between the line $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ and the plane $2x - y + \sqrt{\lambda} z + 4 = 0$ is such that $\sin \theta = \frac{1}{3}$ the value of λ is

(A) $\frac{3}{4}$

(B) $-\frac{4}{3}$

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(C) $\frac{5}{3}$

(D) $-\frac{3}{5}$

Q.18

If the centroid of a tetrahedron $OABC$, where A, B, C are given by $(a, 2, 3), (1, b, 2)$ and $(2, 1, c)$ respectively be $(1, 2, -1)$, then distance of $P(a, b, c)$ from origin is

(A) $\sqrt{107}$

(B) $\sqrt{14}$

(C) $\frac{\sqrt{107}}{14}$

(D) none of these

Q.19

Consider the following statements

(i) Line joining $(1, 2, 5), (4, 3, 2)$ is parallel to the line joining $(5, 1, -11), (8, 2, -8)$.

(ii) Three concurrent lines with D.C's $(l_i, m_i, n_i)\ , i = 1, 2, 3$ are coplanar if $\begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix} = 0$

(iii) The plane $x - 2y + z = 21$ and the line $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-1}{3}$ are parallel.

Which of these is/are correct?

(A) (i) and (ii)

(B) (ii) and (iii)

(C) (iii) and (i)

(D) (i), (ii) and (iii)

Q.20

The locus of a point which moves so that the difference of the squares of its distances from two given points is constant, is a

(A) straight line

(B) plane

(C) sphere

(D) none of these

Q.21

If the planes $x = cy + bz, y = az + cx$ and $z = bx + ay$ pass through a line, then $a^2 + b^2 + c^2 + 2abc$ is equal to

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Q.22

A tetrahedron has vertices at $O(0,0,0), A(1,2,1), B(2,1,3)$ and $C(-1,1,2)$. Then the angle between the faces OAB and ABC will be $\cos^{-1}(p/q)$ where p and q are integers and p is prime then $\mathrm{p}+\mathrm{q}=$

Q.23

The plane $x=0$ divides the join of $(-2,3,4)$ and $(1,-2,3)$ in the ratio $\mathrm{m}:\mathrm{n}$ where m and n are coprime then $\mathrm{m.n}=$

Q.24

If a plane $2x+3y-4z-1+\lambda(3x-y+z+2)=0$ passes through the line of intersection of the planes $2x+3y-4z=1$ and $3x-y+z+2=0$ and makes an intercept of 4 units on the positive x-axis, then $|10\lambda|$ is

Q.25

The distance of the point $(-1,-5,-10)$ from the point of intersection of the line $\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{12}$ and the plane $x-y+z=5$ is

Q.26

The plane $x-y-z-4=0$ is rotated through 90° about its line of intersection with the plane $x+y+2z=4$. Its equation in the new position is $5x+y+4z=k$, then $k=$

Q.27

The line $\frac{x-2}{3}=\frac{y+1}{2}=\frac{z-1}{-1}$ intersects the curve $xy=c^2, z=0$ if $c^2=$

Q.28

The equation of the plane which is parallel to the plane $x-2y+2z=5$ and whose distance from the point $(1,2,3)$ is 1 , is $x-2y+2z=\lambda$ then sum of all possible values of λ is $=$

Q.29

The radius of the circle formed by the sphere $x^2+y^2+z^2=25$ and the plane $x+y+z-3\sqrt{3}=0$ is

Q.30

Projection of the segment AB joining $A(4,7,8)$ and $B(2,3,7)$ on a line whose direction ratios are $-1,2,2$ is p/q where q is prime and p is an integer, then $\mathrm{p}+\mathrm{q}=$

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Answers & Solutions

Q.1 Answer:

$(2, 3, -1)$

Solution:

Let N be the foot of the perpendicular from the point $(0, 2, 3)$ on the given line

$\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$...(i)

Any point on (i) is $(5r - 3, 2r + 1, 3r - 4)$. If this point is NP, then direction ratios of NP are $<$ $5r - 3 - 0, 2r + 1 - 2, 3r - 4 - 3 >$

i.e. $< 5r - 3, 2r - 1, 3r - 7\rangle$. Since PN is $\perp$ to the given line.

$\therefore \quad 5(5r - 3) + 2(2r - 1) + 3(3r - 7) = 0 \Rightarrow 38r - 38 = 0 \Rightarrow r = 1$

$\therefore \quad$ N is $(5 - 3, 2 + 1, 3 - 4)$ i.e. $(2, 3, -1)$

Q.2 Answer:

$x - y + z = 2$

Solution:

Plane contains line (1), so it passes through point $(1, 2, 3)$

$a(x - 1) + b(y - 2) + c(z - 3) = 0$

Now $a(1) + b(3) + c(2) = 0$ ...(i)

as well plane is perpendicular

i.e., $2a + 7b + 5c = 0$ ...(ii)

From equation (i) and (ii) $a : b : c$ is $(1, -1, 1)$ or equation of plane is $(x - 1) - (y - 2) + (z - 3) = 0 \Rightarrow x - y + z = 2$

Q.3 Answer:

$Al + Bm + Cn = 0$

Solution:

The equation of plane containing the line

$\frac{x-x_1}{1} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ is

$a\,(x - x_1) + b\,(y - y_1) + c\,(z - z_1) = 0$

Since, the normal to the plane is perpendicular to the given line.

Therefore, $al + bm + cn = 0$

Q.4 Answer:

$\vec{a}.\vec{c} = \vec{b} \cdot \vec{c}$

Solution:

The lines $\mathrm{r} = \mathrm{a} + \lambda(\mathbf{b} \times \mathrm{c})$ and $\mathrm{r} = \mathbf{b} + \mu(\mathrm{c} \times$ a) pass through points a and b respectively and are parallel to vector $\mathrm{b} \times \mathrm{c}$ and $\mathrm{a} \times \mathbf{c}$ respectively. Therefore, they intersect if $\mathrm{a} - \mathrm{b}, \mathrm{b} \times \mathrm{c}$

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and c × a are coplanar and so

$(a-b)\cdot\{(b\times c)\times(c\times a)\}=0$

$\Rightarrow (\mathrm{a}-\mathrm{b})\cdot\{[\mathrm{bca}]\mathrm{c}-[\mathrm{bcc}]\mathrm{a}\}=0$

$\Rightarrow ((a-b)\cdot c)[bca]=0$

$\Rightarrow$ **a.c − b.c** $=0\Rightarrow$ **a.c = b.c**

Q.5 Answer:

$\frac{l_1+l_2}{2\cos\theta/2}, \frac{m_1+m_2}{2\cos\theta/2}, \frac{n_1+n_2}{2\cos\theta/2}$

Solution:

Let OA and OB be two lines with DCs l_1, m_1, n_1 and l_2, m_2, n_2. Let $OA = OB = 1$. Then the coordinates of A and B are (l_1, m_1, n_1) and (l_2, m_2, n_2) respectively. Let OC be the bisector of $\angle AOB$. Then C is the mid point of AB and so its coordinates are

$\left(\frac{l_1+l_2}{2}, \frac{m_1+m_2}{2}+\frac{n_1+n_2}{2}\right)$

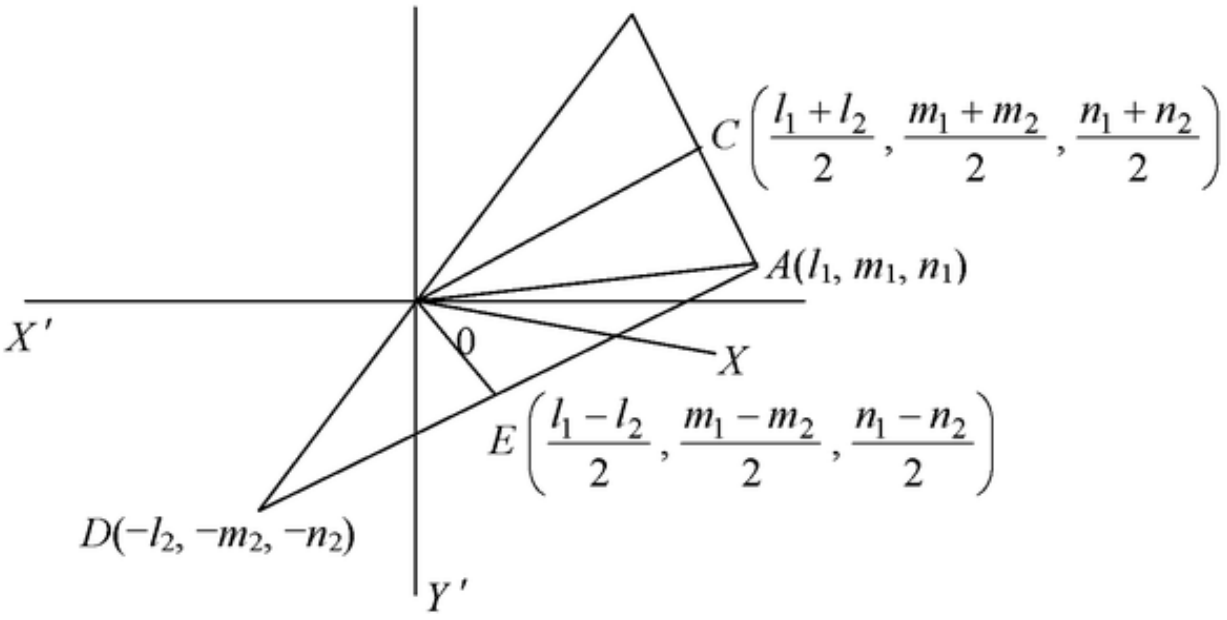
DRs of $\frac{l_1+l_2}{2}, \frac{m_1+m_2}{2}, \frac{n_1+n_2}{2}$

We have $OC=\sqrt{\left(\frac{l_1+l_2}{2}\right)^2+\left(\frac{m_1+m_2}{2}\right)^2+\left(\frac{n_1+n_2}{2}\right)^2}$

$=\frac{1}{2}\sqrt{(l_1^2+m_1^2+n_1^2)+(l_2^2+m_2^2+n_2^2)+2(l_1l_2+m_1m_2+n_1n_2}$

$=\frac{1}{2}\sqrt{2+2\cos\theta}$

$=\frac{1}{2}\sqrt{2(1+\cos\theta)}=\cos\left(\frac{\theta}{2}\right)$



$\therefore$ **DCs of** $\overrightarrow{OC}$ **are** $\frac{l_1+l_2}{2(OC)}, \frac{m_1+m_2}{2(OC)}, \frac{n_1+n_2}{2(OC)}$

$\frac{l_1+l_2}{2\cos\theta/2}, \frac{m_1+m_2}{2\cos\theta/2}, \frac{n_1+n_2}{2\cos\theta/2}$

Q.6 Answer:

$\frac{6}{\sqrt{14}}$

Solution:

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The line can be as well written as $3\overrightarrow{\mathrm{i}}-\overrightarrow{\mathrm{j}}+\mathrm{t}(\overrightarrow{\mathbf{i}}+2\overrightarrow{\mathbf{j}}+3\overrightarrow{\mathbf{k}})$

Now the projection of $\overrightarrow{\mathrm{i}}+\overrightarrow{\mathrm{j}}+\overrightarrow{\mathrm{k}}$ on the line is given by the projection onto the direction vector, i.e. $\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$ in general is the projection of a on $\vec{b}$.

Here, its $\frac{1+2+3}{\sqrt{1+4+9}}=\frac{6}{14}$

Q.7 Answer:

$\frac{2}{3}$

Solution:

Solving the planes taking 3 at a time

Vertices of tetrahedron are

$O:(0,0,0), A:(1,1,-1), B:(1,-1,1), C:(-1,1,1)$

$\Rightarrow$ Volume $=\frac{1}{3}\times$ (area of base) $\times$ (height) $=\frac{1}{3}\times\frac{\sqrt{3}}{4}\times(2\sqrt{2})^2\times\frac{1}{\sqrt{3}}=\frac{2}{3}$

Or

Volume $=\frac{1}{6}\left[\begin{array}{ccc}\vec{a}\vec{b} & \vec{c}\end{array}\right]$ (where $\vec{a}=OA, \vec{b}=OB, \vec{c}=OC$)

Q.8 Answer:

$\frac{1}{\sqrt{3}}$

Solution:

Vector normal to the planes $2x+3y+z=1$ is $\vec{a}=2\hat{i}+3\hat{j}+\hat{k}$

Vector normal to the planes $x+3y+2z=2$ is $\vec{b}=\hat{i}+3\hat{j}+2\hat{k}$

Vector parallel to line of intersection of given planes $=\vec{a}\times\vec{b}=3\hat{i}-3\hat{j}+3\hat{k}$

Angle α between $3\hat{i}-3\hat{j}+3\hat{k}$ and $\hat{i}$ is given by $\cos\alpha=\frac{1}{\sqrt{3}}$

Q.9 Answer:

$(-1,2,-2)$

Solution:

$(-1,2,-2)$

Let the coordinates of P be $(\mathrm{x},\mathrm{y},\mathrm{z})$. Then,

Direction ratios of $\mathrm{OP}=$ Coordinates of P - Coordinates of O $-1,2,2=(x-0),(y-0),(z-0)$

Thus, coordinates of P are $(-1,2,-2)$.

Q.10 Answer:

(a,b,c)

Solution:

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Intercepts are $(3a, 0, 0), (0, 3a, 0), (0, 0, 3a)$ hence option D. So, the plane cuts x-axis at $(3a, 0, 0)$, so x-intercept $= 3a$
The plane cuts y-axis at $(0, 3b, 0) \Rightarrow$ y-intercept $= 3b$.
The plane cuts z-axis at $(0, 0, 3c) \Rightarrow$ z-intercept $= 3c$

Q.11 Answer:

intersect in a line

Solution:

All the planes pass through the point $(-d, 0, 0)$. Also $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix} = 0$

$\Rightarrow$ the three planes intersect in a line.

Q.12 Answer:

1

Solution:

Direction cosines of line $= \left(\frac{2}{7}, \frac{3}{7}, \frac{-6}{7}\right)$
Now, $x' = 1 + \frac{2r}{7}, y' = -2 + \frac{3r}{7}$ **and** $z' = 3 - \frac{6r}{7}$
$\therefore \left(1 + \frac{2r}{7}\right) - \left(-2 + \frac{3r}{7}\right) + \left(3 - \frac{6r}{7}\right) = 5 \Rightarrow r = 1$

Q.13 Answer:

$3x + 4y - 5z = 9$

Solution:

The equation of a plane passing $(2, 2, 1)$ is
$a(x-2) + b(y-2) + c(z-1) = 0$
This passes though $(9, 3, 6)$ and is $\perp$ to the plane $2x + 6y + 6z - 1 = 0$
$\therefore \quad 7a + 1 \cdot b + 5c = 0 \quad$ **and** $\quad 2a + 6b + 6c = 0$
$= \frac{a}{-24} = \frac{b}{-32} = \frac{c}{40}$
$\Rightarrow \frac{a}{3} = \frac{b}{4} = \frac{c}{-5}$
So, the required plane is $3(x-2) + 4(y-2) - 5(z-1) = 0$
or $3x + 4y - 5z = 9$

Q.14 Answer:

$x - y + z = 0$

Solution:

Let coordinate of C be (α, β, γ) As $AC = BC \Rightarrow (\alpha - 1)^2 + (\beta + 1)^2 + (\gamma - 1)^2 = (\alpha + 1)^2 + (\beta - 1)^2 + (\gamma + 1)^2$
$\Rightarrow$ locus of C is $x - y + z = 0$

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Q.15 Answer:

$\cos^{-1}\frac{1}{3}$

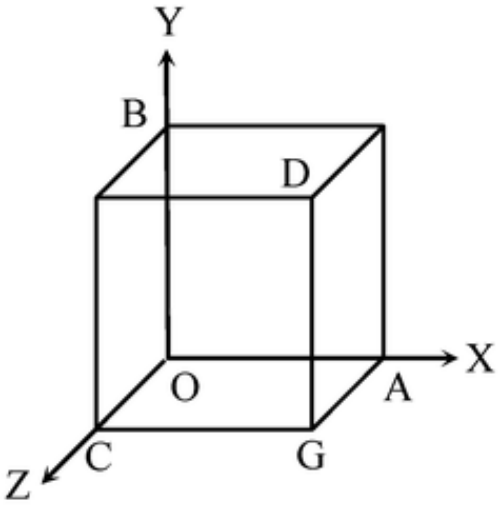
Solution:

Let the side of the cube be ' a '.

Let $O(0,0,0), D(a,a,a), B(0,a,0), G(a,0,a)$

drs of OD **are** (a,a,a) **and drs of** BG **are** $(a,-a,a)$

Hence, angle between OD **and** BG **is** $\cos^{-1}\left(\frac{a^2-a^2+a^2}{\sqrt{3a^2}\cdot\sqrt{3a^2}}\right)=\cos^{-1}\left(\frac{1}{3}\right)$



Q.16 Answer:

$\vec{r}\cdot(4\hat{i}-12\hat{j}-3\hat{k})=32$

Solution:

The equation of a plane parallel to the plane $\vec{r}\cdot(4\hat{i}-12\hat{j}-3\hat{k})-7=0$ **is**

$\vec{r}.(4\hat{i}-12\hat{j}-3\hat{k})+\lambda=0$. **This passes through** $2\hat{i}-\hat{j}-4\hat{k}$. **Therefore**

$(2\hat{i}-\hat{j}-4\hat{k})\cdot(4\hat{i}-12\hat{j}-3\hat{k})+\lambda=0$

$\Rightarrow \quad 8+12+12+\lambda=0\Rightarrow\lambda=-32$

So, the required plane is $\vec{r}\cdot(4\hat{i}-12\hat{j}-3\hat{k})=32$

Q.17 Answer:

$\frac{5}{3}$

Solution:

Angle between line and normal to plane is

$\cos\left(\frac{\pi}{2}-\theta\right)=\frac{1\times2+2\times(-1)+2\sqrt{\lambda}}{3\times\sqrt{5+\lambda}}$

where θ **is angle between line and plane**

$\Rightarrow \quad \sin\theta=\frac{1\times2+2\times(-1)+2\sqrt{\lambda}}{3\times\sqrt{5+\lambda}}$

$\Rightarrow \quad \frac{1}{3}=\frac{2\sqrt{\lambda}}{3\sqrt{5+\lambda}}$

$\Rightarrow \quad \lambda=\frac{5}{3}$

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Q.18 Answer:

$\sqrt{107}$

Solution:

Centriod is $\left(\frac{1}{4}\sum x, \frac{1}{4}\sum y, \frac{1}{4}\sum z\right) = (1, 2, -1)$

$\Rightarrow \quad \frac{a+1+2+0}{4} = 1, \frac{2+b+1+0}{4} = 2, \frac{3+2+c+0}{4} = -1$

$\Rightarrow \quad a = 1, b = 5, c = -9 \quad \therefore \quad OP = \sqrt{a^2 + b^2 + c^2} = \sqrt{107}$

Q.19 Answer:

(ii) and (iii)

Solution:

(i) $\frac{4-1}{8-5} = \frac{3-2}{2-1} \neq \frac{2-5}{-8+11}$

$\therefore$ **two lines are not parallel.**

(ii) If the vectors $(li)\hat{i} + (mi)\hat{j} + (ni)\hat{k}, i = 1, 2, 3$ **are coplanar then the lines are also coplanar as the lines pass through the same point. These vectors coplanar if**

$$\begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix} = 0$$

(iii) $1(1) - 2(2) + 3(1) = 0$

$\therefore$ **given line and plane all parallel.**

Q.20 Answer:

plane

Solution:

Let the two fixed points be $(\mathrm{x}_1, \mathrm{y}_1)$ **and** $(\mathrm{x}_2, \mathrm{y}_2)$

Given that $(\mathrm{x} - \mathrm{x}_1)^2 + (\mathrm{y} - \mathrm{y}_1)^2 + (\mathrm{z} - \mathrm{z}_1)^2 - \left[(\mathrm{x} - \mathrm{x}_2)^2 + (\mathrm{y} - \mathrm{y}_2)^2 + (\mathrm{z} - \mathrm{z}_2)^2\right] = \mathrm{k}$

$\Rightarrow 2\mathrm{x}(\mathrm{x}_1 + \mathrm{x}_2) + 2\mathrm{y}(\mathrm{y}_1 + \mathrm{y}_2) + 2\mathrm{z}(\mathrm{z}_1 + \mathrm{z}_2) + \mathrm{k} -$
$\left(\mathrm{x}_1^2 + \mathrm{y}_1^2 + \mathrm{z}_1^2 + \mathrm{x}_2^2 + {\mathrm{y}_2}^2 + {\mathrm{z}_2}^2\right) =$**0**

Therefore, locus is plane.

Q.21 Answer:

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Solution:

Let l, m, n **be the direction ratios of the line lying in the three planes, then the line is perpendicular to the normals to these planes, so that we have**

$l - cm - bn = 0$

$-cl + m - an = 0$

$-bl - am + n = 0$

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Eliminating l, m, n from these equations we $\begin{vmatrix} 1 & -c & -b \\ -c & 1 & -a \\ -b & -a & 1 \end{vmatrix} = 0$

$\Rightarrow \quad 1 - a^2 + c(-c - ab) - b(ac + b) = 0 \Rightarrow a^2 + b^2 + c^2 + 2abc = 1$

Q.22 Answer:

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Solution:

Angle between two plane faces is equal to the angle between the normals n_1 and n_2 to the planes.

n_1 the normal of face OAB is given by $\overrightarrow{OA} \times \overrightarrow{OB} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 2 & 1 & 3 \end{vmatrix} = 5\hat{i} - \hat{j} - 3\hat{k}$ **...(i)**

n_2 the normal of face ABC is given by $\overrightarrow{AB} \times \overrightarrow{AC}$

$2 - 1, 1 - 2, 3 - 1$ **and** $-1 - 1, 1 - 2, 2 - 1$ **i.e.,** $1, -2, 2$ **and** $-2, -1, 1$

$\therefore n_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ -2 & -1 & 1 \end{vmatrix} = \hat{i} - 5\hat{j} - 3\hat{k}$ **...(ii)**

If θ be the angle between n_1 and n_2, then $\cos\theta = \frac{n_1 \cdot n_2}{|n_1| \cdot |n_2|} = \frac{5+5+9}{\sqrt{35} \cdot \sqrt{35}} = \frac{19}{35} \quad \Rightarrow = \cos^{-1}\left(\frac{19}{35}\right)$

Q.23 Answer:

—

Solution:

Any point in the plane $x = 0$ is $(0, \beta, \gamma)$. Now let the ratio is $\lambda : 1$, then

$\frac{\lambda(1)+1(-2)}{\lambda+1} = 0$ **(x co-ordinate)**

$\Rightarrow \lambda = \frac{2}{1} = 2$

Q.24 Answer:

—

Solution:

$(2 + 3\lambda)x + (3 - \lambda)y + (\lambda - 4)z = 1 - 2\lambda$ **makes intercept on x-axis 4.**

$\therefore \quad \frac{1-2\lambda}{2+3\lambda} = 4 \quad \Rightarrow \quad \lambda = -\frac{1}{2}$

Q.25 Answer:

—

Solution:

Any point on the line $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12} = t$ **is** $(3t + 2, 4t - 1, 12t + 2)$

This lies on $x - y + z = 5$

$\therefore \quad 3t + 2 - 4t + 1 + 12t + 2 = 5$ i.e., $11t = 0 \Rightarrow t = 0$

$\therefore \quad$ **Point is** $(2, -1, 2)$

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Its distance from $(-1, -5, -10)$ is

$= \sqrt{(2+1)^2 + (-1+5)^2 + (2+10)^2} = \sqrt{9+16+144} = 13$

Q.26 Answer:

—

Solution:

Equation of family of planes passing through line of intersection of the two given planes is $x - y - z - 4 + \lambda(x + y + 2z - 4) = 0$

i.e. $(1+\lambda)x + (\lambda - 1)y + (2\lambda - 1)z - 4(\lambda + 1) = 0$

If this plane is perpendicular to $x - y - z - 4 = 0$, then

(1) $(1+\lambda) + (-1)(\lambda - 1) + (-1)(2\lambda - 1) = 0$

$1 + \lambda - \lambda + 1 - 2\lambda + 1 = 0$

$3 - 2\lambda = 0$

$\Rightarrow \quad \lambda = \frac{3}{2} \Rightarrow$ Equation of required plane is $\frac{5}{2}x + \frac{1}{2}y + 2z - 10 = 0$

$\Rightarrow \quad 5x + y + 4z = 20$

Q.27 Answer:

—

Solution:

At the point on the lien where it intersects the curve, we have $z = 0$, so that

$$\frac{x-2}{3} = \frac{y+1}{2} = \frac{0-1}{-1}$$

$$\Rightarrow \frac{x-2}{3} = 1 \frac{y+1}{2} = 1$$

$\Rightarrow \quad x = 5$ and $y = 1$.

Putting these values of x and y in $xy = c^2$, we get $c^2 = 5$.

Q.28 Answer:

—

Solution:

Equation of plane parallel to $x - 2y + 2z = 5$ is $x - 2y + 2z + k = 0$

Now, according to equation $\frac{1-4+6+k}{\sqrt{9}} = \pm 1 \quad$ or $k + 3 = \pm 3$

Now, according to equation
$\frac{1-4+6+k}{\sqrt{9}} = \pm 1 \quad$ or $k + 3 = \pm 3$

$\Rightarrow \quad k = 0$ or-6;

$\therefore \quad x - 2y + 2z - 6 = 0$ or $x - 2y + 2z = 6$

Q.29 Answer:

—

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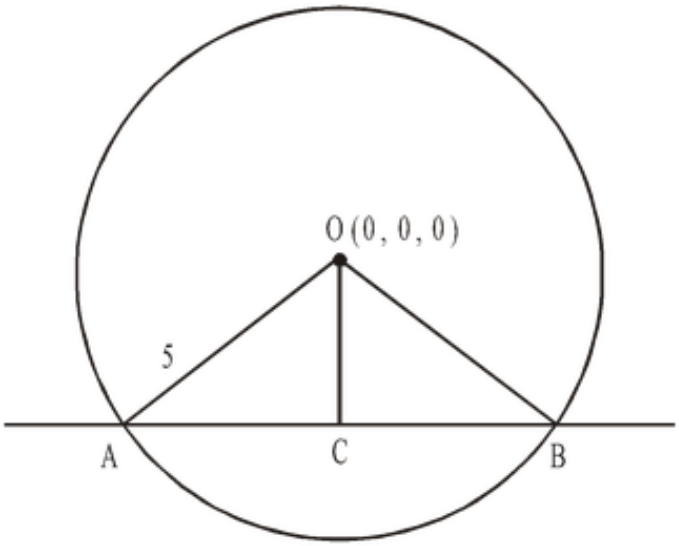
Solution:

Let $O(0,0,0)$ be centre of the sphere and C is the centre of the circle

Now, $OC = \left|\frac{0+0+0-3\sqrt{3}}{\sqrt{3}}\right| = 3$

Therefore radius of circle

$= AC = \sqrt{OA^2 - OC^2} = \sqrt{25-9} = 4$



Q.30 Answer:

—

Solution:

Projection length $= \frac{2(-1)}{3} + \frac{4.2}{3} + \frac{1.2}{3} = \frac{8}{3}$

(formula : $(|(x_1 - x_2)\, l + (y_1 - y_2)\, m + (z_1 - z_2)\, n|)$

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